



Project page

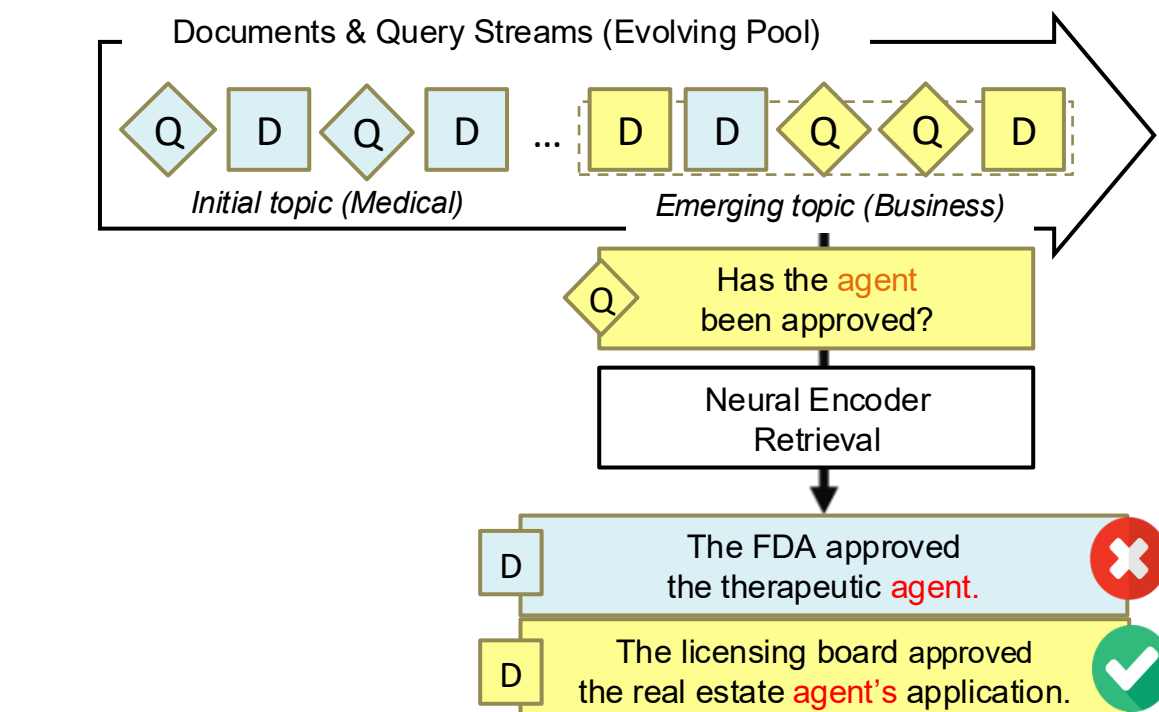


Source code

1. Motivation

Continual Retrieval (CR)

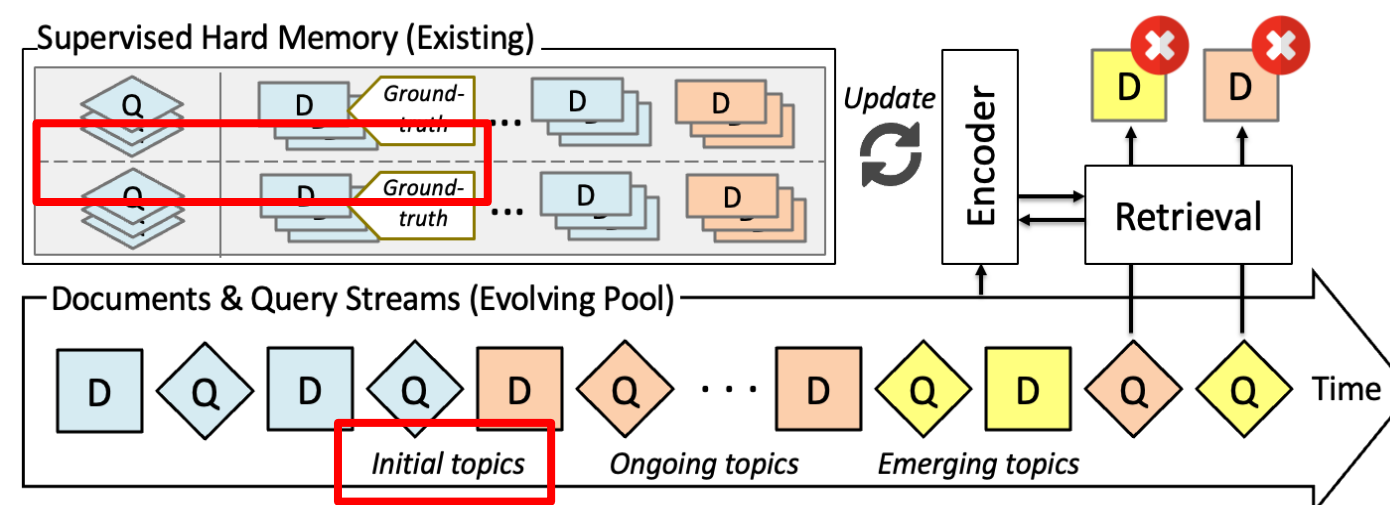
Continual learning on dense retrieval in evolving data streams.



Limitations of Existing Methods for CR

Fixed & Past Labeled Pairs

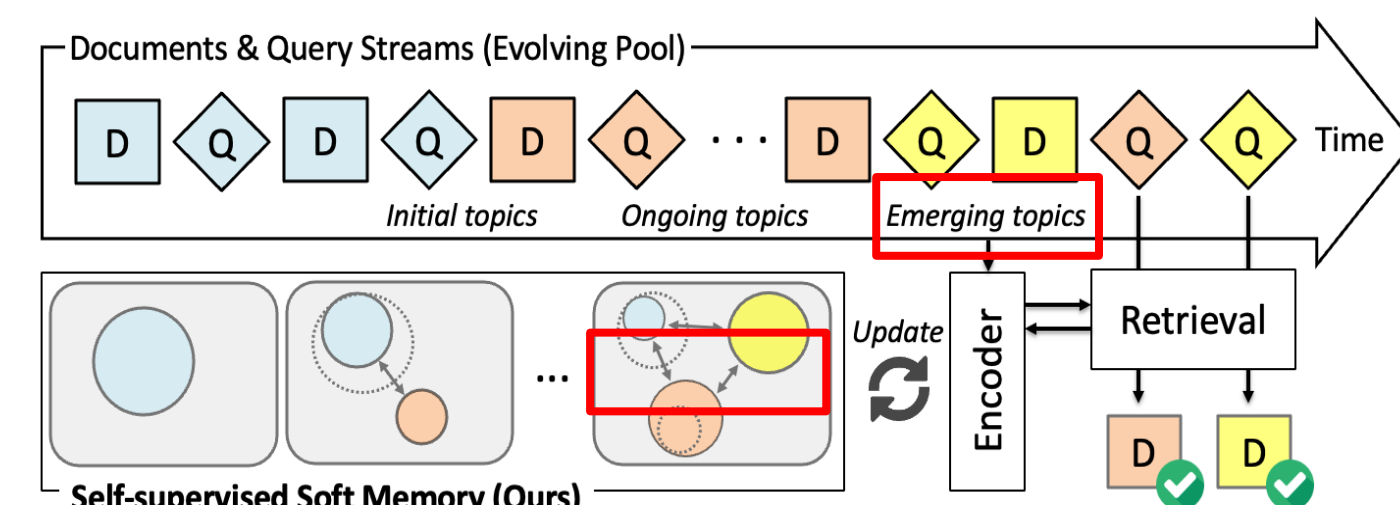
- No generalization to unseen queries.
- Requires ground-truth labels.



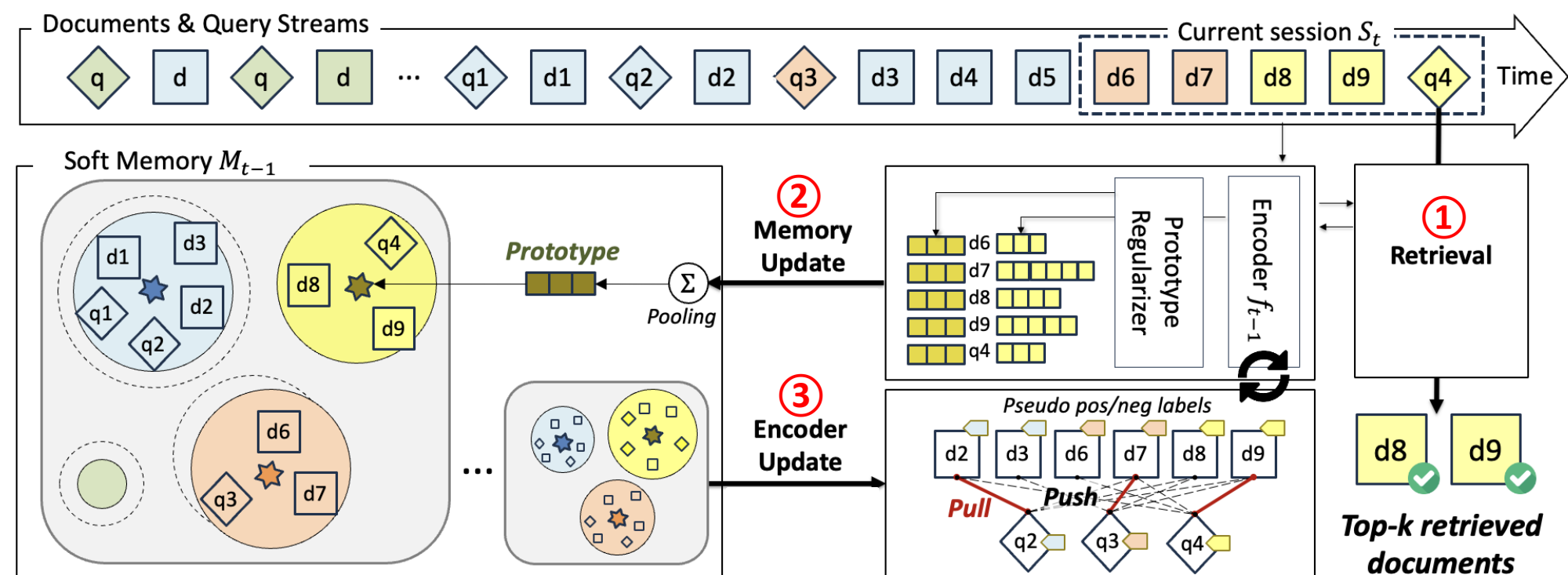
Soft Memory

Dynamic Topical-Group-Based Pairs

- Generalization to unseen queries.
- Requires no labels.



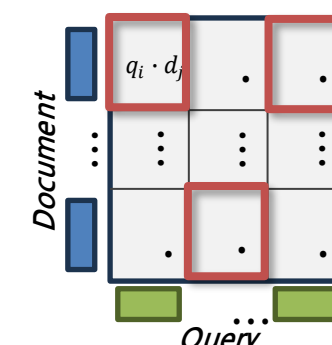
2. CREAM: Soft Memory-Based Self-Supervised Learning



① Token-Level Similarity

To Leverage rich context in label-free settings.

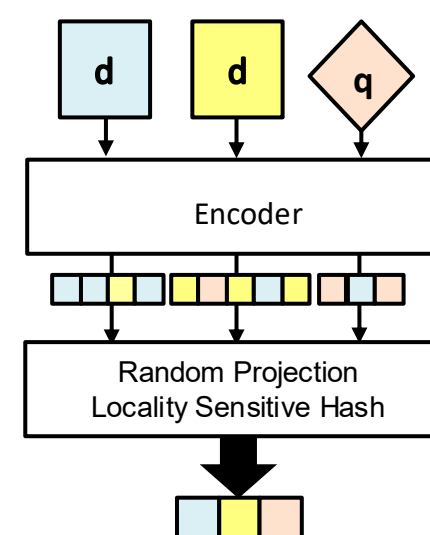
$$Sim_{qd} = \sum_{i \in [E_q]} \max_{j \in [E_d]} E_{q_i} \cdot E_{d_j}^T$$



② Streaming Clustering based on Token-Level Similarity

To efficiently perform computationally expensive operations online.

- ✓ Fixed-length prototypes from variable-length token arrays.
- ✓ Incrementally maintained statistics for adaptive decay of outdated clusters.



③ Contrastive Learning with Coreset Pairs

To maximize coverage while minimizing redundancy.

3. Experiments & Results

Supervised Results

Model	LoTTE				MSMARCO			
	Disjoint (Avg.)		Shared (Avg.)		Disjoint (Avg.)		Shared (Avg.)	
	S@5	R@10	S@5	R@10	S@5	R@10	S@5	R@10
ColBERT ⁺⁺	42.13	23.30	21.23	10.00	78.29	82.95	47.76	51.86
ER [*]	67.59	43.24	45.93	25.06	96.32	97.69	76.30	80.85
MIR [*]	67.33	42.57	45.61	24.87	95.66	96.90	<u>76.37</u>	<u>81.08</u>
GSS [*]	68.15	43.57	44.80	24.14	95.77	94.49	75.82	80.63
OCS [*]	66.77	42.75	44.07	23.76	95.98	97.14	73.07	77.86
L2R [*]	55.06	31.67	32.80	16.29	91.00	94.04	62.98	68.58
CREAM	68.20	43.57	48.60	27.11	93.15	95.15	73.68	77.89

Unsupervised Results

Model	LoTTE				MSMARCO			
	Disjoint (Avg.)		Shared (Avg.)		Disjoint (Avg.)		Shared (Avg.)	
	S@5	R@10	S@5	R@10	S@5	R@10	S@5	R@10
BM25	59.89	33.20	40.77	19.88	72.79	75.49	50.62	54.54
ColBERT ⁺	4.11	1.43	3.70	2.04	41.19	51.50	13.92	17.43
ER	40.51	25.73	22.61	10.76	75.92	81.86	41.60	47.06
MIR	31.83	16.69	19.55	8.97	76.44	82.32	44.12	49.73
GSS	37.25	20.53	21.08	9.86	75.33	82.29	37.60	43.16
OCS	39.43	22.24	20.33	9.64	73.52	79.89	42.63	48.68
L2R	41.57	23.60	22.61	9.92	75.05	81.44	36.82	42.60
CREAM	68.20	43.57	48.60	27.11	93.15	95.15	73.68	77.89

Experimental Settings

- Streaming sessions: LoTTE, MSMARCO(5 domain, QA) Datasets.
- Evaluation Metrics: Success@5, Recall@10.
- **Bold**: Best performance in unsupervised settings.
- Underline: Best performance in all settings including supervised.

Highlights

- CREAM achieves the best performance in the unsupervised setting.
- CREAM even outperforms supervised methods on LoTTE.